Intro to ROS

Autonomous Car, Group 5

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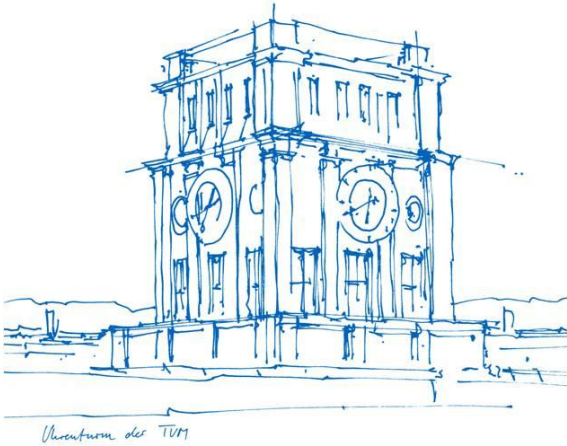
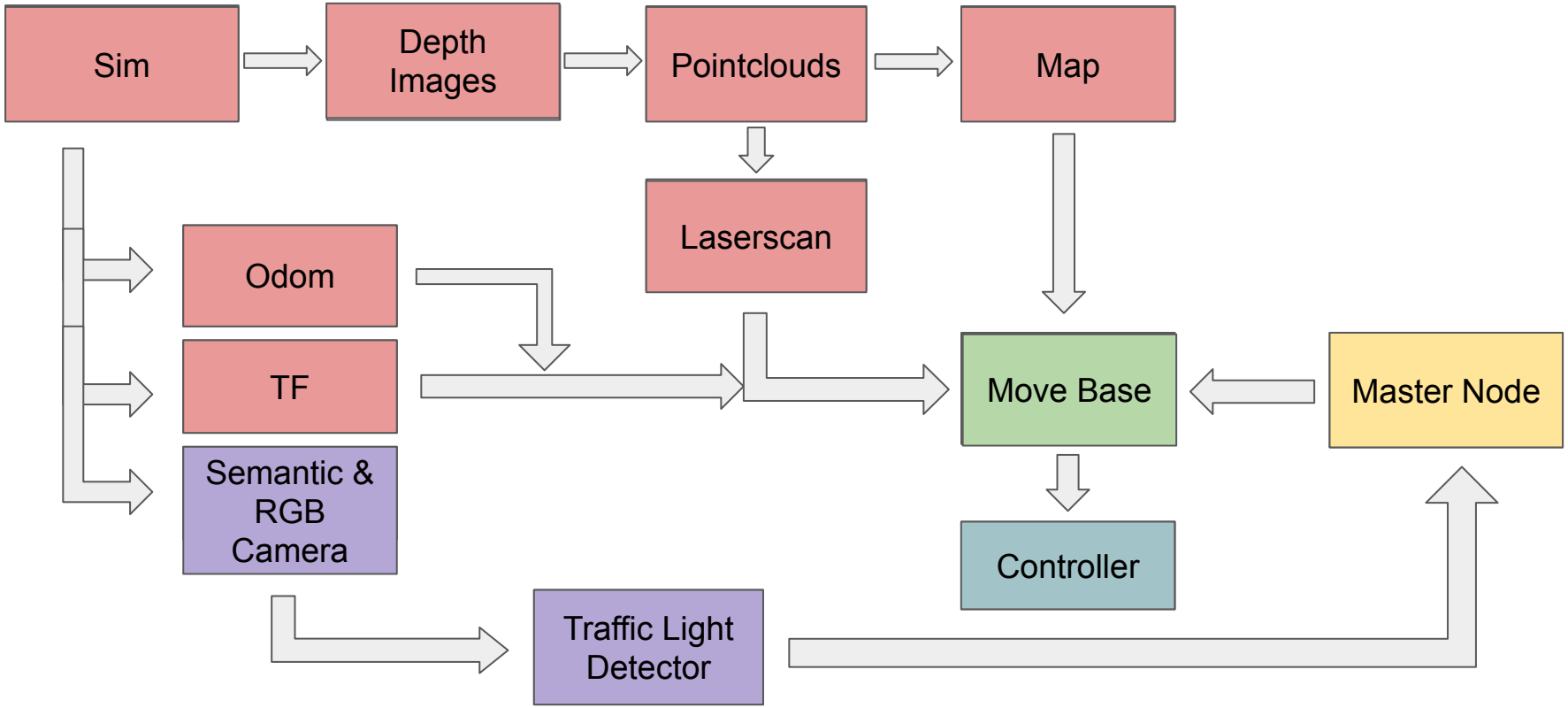
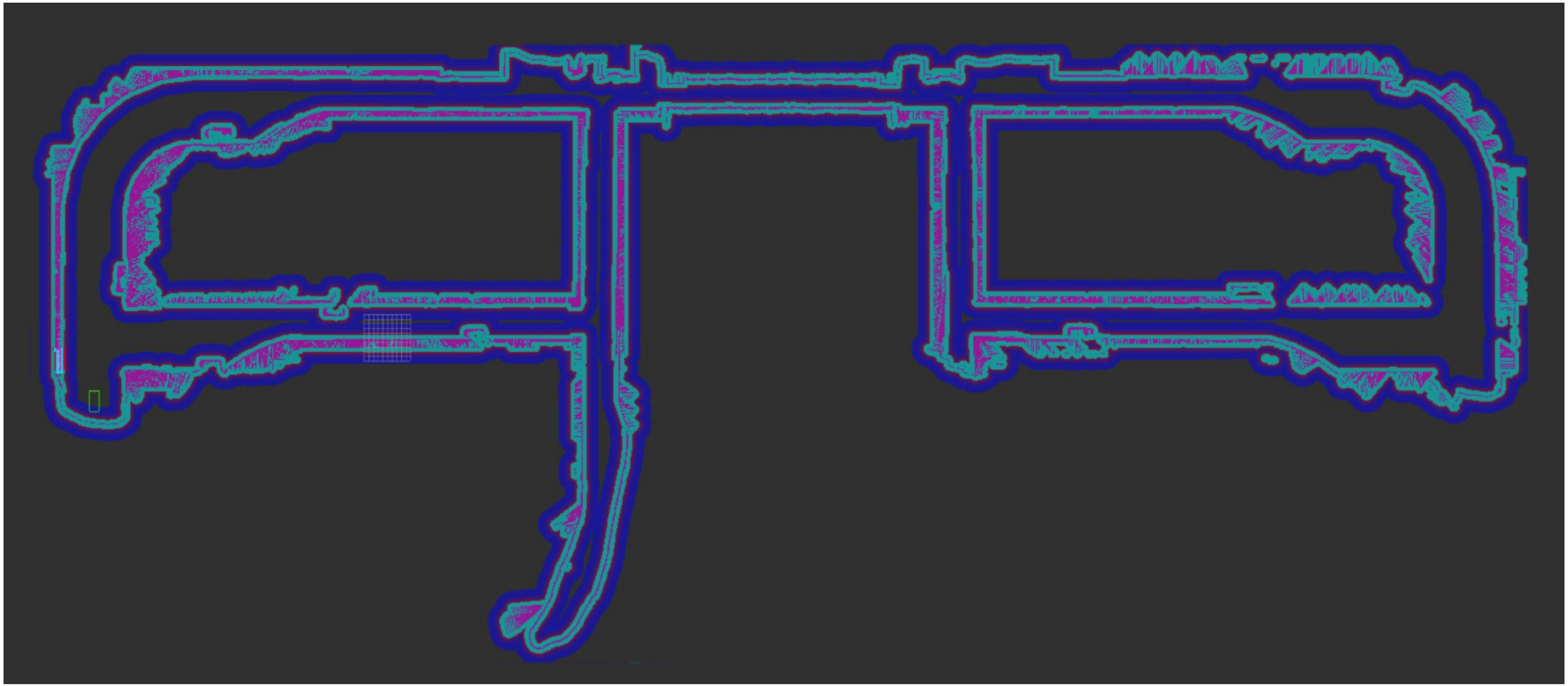


ROS Graph (simplified)



Global costmap generated by move_base



ROS Packages Used

Table 1: External code used in the project

Name	Description	Ref.
depth_image_proc	Converts the depth image to a point cloud	[5]
octomap_server	Used to map the environment	[6]
pointcloud_to_laserscan	Converts a Pointcloud into a 2D Laserscan	[7]
map_server	Used to load and publish a saved occupancy map of the environment	[2]
move_base	Provides a ROS interface for configuring, running, and interacting with the navigation stack	[1]
navfn	Global planner that uses the Dijkstra algorithm	[3]
teb_local_planner	Optimization based local planner that plans the trajectory	[4]
cv_bridge	Acts as a bridge between ROS image message and OpenCV	[8]
image_transport	For image transportation and compression	[9]

Difficulties

- Frame / TF issues in the beginning
- Tried to use move_base without global map at first, but hard and not well documented
- Get teb_local_planner to plan fast enough
- Accurate calculation from twist_msg to the actuator_msg in the controller